Bayesian Cox Proportional Hazards Model for analyzing microbiome data

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Abstract

Compositional covariates are common in survival analysis, particularly in microbiome research, where their unit-sum constraint induces severe collinearity. This collinearity can compromise coefficient interpretability and the reliability of statistical inference in Cox proportional hazards regression. To address these challenges, we propose a Bayesian Cox modeling framework that integrates structured coefficient constraints and flexible sparsity-inducing priors. The framework incorporates three methodological innovations. First, a soft sum-to-zero constraint is imposed on regression coefficients, alleviating collinearity while preserving their interpretation as log-fold changes relative to the geometric mean abundance. Second, we adapt the global shrinkage hyperparameter of the regularized horseshoe prior to the Cox model using the effective number of uncensored events, enabling robust performance across varying sparsity levels. Third, to account for phylogenetic relationships in microbiome features, we introduce a conditional autoregressive (CAR) prior on the local shrinkage parameters, allowing correlated shrinkage among coefficients of evolutionarily related taxa. Extensive simulations demonstrate that our framework achieves improved predictive accuracy and variable selection compared to standard penalized methods. Application to a type 1 diabetes microbiome dataset further highlights its utility in identifying microbial signatures associated with disease onset.

Keywords: Bayesian survival analysis; Compositional data; Variable selection; Sparse priors; Cox proportional hazards model.

1 Introduction

The human microbiome plays a vital role in health, influencing processes such as human development, nutrient absorption, immune regulation, and disease resistance. Increasing evidence links microbiome composition to a variety of diseases and physiological states [1, 2, 3, 4]. Over the past decade, advances in high-throughput sequencing technologies—such as 16S ribosomal RNA (16S rRNA) gene sequencing and taxonomic profiling—have generated large-scale microbiome datasets. These data have motivated extensive statistical research spanning diverse outcome types, including continuous, binary, survival, and competing risks [5, 6].

Despite these advances, microbiome data present notable statistical challenges. First, although OTU counts are recorded as absolute numbers, biological interpretation relies on relative abundances constrained to sum to one. This compositional property induces perfect collinearity: changes in one taxon necessarily alter all others, complicating regression inference. Second, in survival analysis, these challenges are compounded when high-dimensional microbiome profiles must be linked to complex time-to-event outcomes. Third, sequencing data often exhibit phylogenetic structure—reflecting evolutionary relationships among taxa—that introduces dependence among covariates and should be formally incorporated into models [4, 7].

Several techniques for compositional data analysis (CoDA) exist, such as LEfSe [8], metagenomeSeq [9], ANCOM-II [10], ANCOM-BC [11], ALDEx2 [12], and corncob [13]. These methods are primarily designed for cross-sectional abundance comparisons and do not generalize to time-to-event outcomes, limiting their utility in survival studies.

For microbiome survival analysis, most approaches either ignore compositional constraints or apply additive log-ratio (ALR) [14, 15] or centered log-ratio (CLR) [14] transformations prior to fitting traditional survival models. McGregor et al. (2020) [16] introduced a CoDA survival regression using the isometric log-ratio (ILR) transformation within the Cox model. While theoretically sound, ILR is challenging to implement and interpret in high dimensions. Pujolassos et al. (2024) [17] developed coda4microbiome, combining all pairwise log-ratios with penalized Cox regression. Reparameterization improves interpretability, but the pairwise transformation dramatically inflates dimensionality, increasing computation and estimation burden. Fei et al. (2024) [18] proposed FLORAL, a scalable log-ratio LASSO method with augmented Lagrangian zero-sum

constraints, applicable to continuous, binary, survival, and competing risks, including longitudinal time-dependent covariates. However, FLORAL does not account for phylogenetic dependencies.

Bayesian approaches have also been eplored. Zhang et al. (2021) [19] applied z-priors for compositional constraints and Ising priors for feature relationships. Zhang et al. (2024) [4] introduced the Bayesian compositional generalized linear model (BCGLM), log-transforming covariates and enforcing a soft sum-to-zero ("soft-centers") constraint [20] on regression coefficients. BCGLM combined a regularized horseshoe prior with a conditional autoregressive (CAR) structure on local shrinkage parameters to model phylogenetic association. While effective for generalized linear models, BCGLM has not yet been extended to the Cox proportional hazards setting.

To bridge these gaps, we propose a Bayesian Cox proportional hazards model tailored for high-dimensional compositional microbiome data. Our framework introduces: Soft sum-to-zero constraints on regression coefficients for valid interpretation under compositionality. Multiple shrinkage priors—horseshoe, regularized horseshoe, Bayesian LASSO, and Spike-and-Slab LASSO—incorporated within the Cox model without additional covariates or complex reparameterization as required by coda4microbiome. Cox-specific adaptation of the regularized horseshoe global shrinkage hyperparameter based on the effective number of events, improving robustness across sparsity levels. Phylogenetically informed CAR priors on local shrinkage parameters to capture correlation among evolutionarily related taxa. We evaluate the proposed method through extensive simulations under diverse conditions and demonstrate its practical utility in identifying type 1 diabetes (T1D)-associated microbial signatures in a mouse gut microbiome dataset.

The remainder of this paper is organized as follows: Section 2 details the model specification, prior settings, and posterior inference procedures; Section 3 presents simulation results; Section 4 analyzes the T1D dataset; and Section 5 discusses findings and potential extensions. Additional results are listed in the Appendix.

2 Methods

We consider survival data from n subjects. For subject i ($i = 1, \ldots, n$), let U_i denote the event time and C_i the right-censoring time. The observed time is $t_i = \min\{U_i, C_i\}$, with censoring indicator $\delta_i = \mathbf{1}\{U_i \leq C_i\}$. Let $\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ip})^\top$ be the compositional vector of p taxa for subject i, satisfying $x_{ij} \geq 0$ and $\sum_{j=1}^{p} x_{ij} = 1$. The full dataset is denoted $\mathcal{O} = \{(t_i, \delta_i, \boldsymbol{x}_i) : i = 1, \ldots, n\}$.

Cox model with compositional covariates

Our objective is to model the association between $\boldsymbol{x}_i$ and the hazard of the event. The Cox proportional hazards model [21] specifies

$$\lambda(t \mid \boldsymbol{x}_i) = \lambda_0(t) \exp\{\boldsymbol{x}_i^\top \boldsymbol{\beta}\}, \quad i = 1, \ldots, n,$$

where $\lambda_0(t)$ is the baseline hazard function and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^\top$ are the log-hazard ratios.

Because $\boldsymbol{x}_i$ are compositional (unit-sum constraint), a log-ratio transformation is applied. Following Aitchison [14, 15], the additive log-ratio (ALR) transform with the p-th component as reference is $\boldsymbol{x}_i^* = (\log(x_{i1}/x_{ip}), \ldots, \log(x_{i,p-1}/x_{ip}))^\top$, leading to $\lambda(t \mid \boldsymbol{x}_i) = \lambda_0(t) \exp\left\{\boldsymbol{x}_i^{*\top} \boldsymbol{\beta}_{-p}^*\right\}$, where $\boldsymbol{\beta}_{-p}^* = (\beta_1^*, \ldots, \beta_{p-1}^*)^\top$.

This is equivalent to using the elementwise log transform

$$\lambda(t \mid \boldsymbol{x}_i) = \lambda_0(t) \exp\{\boldsymbol{z}_i^\top \boldsymbol{\beta}^*\},$$

where $\boldsymbol{z}_i = (\log(x_{i1}), \ldots, \log(x_{ip}))^\top$, $\boldsymbol{\beta}^* = (\beta_1^*, \ldots, \beta_p^*)^\top$ and $\beta_p^* = -\sum_{j=1}^{p-1} \beta_j^*$, enforcing $\sum_{j=1}^{p} \beta_j^* = 0$. We impose this zero-sum constraint softly via [20]:

$$\sum_{j=1}^{p} \beta_j^* \;\sim\; N\big(0,\; (0.001\, p)^2\big)\,, \tag{1}$$

which is known as "soft-centers".

Defining the linear predictor

$$\eta_i = \log \frac{\lambda(t \mid \boldsymbol{x}_i)}{\lambda_0(t)} = \boldsymbol{z}_i^\top \boldsymbol{\beta}^*,$$

we see the model is a generalized linear form. Inference is based on the Cox partial likelihood: $\mathcal{L}(\boldsymbol{\beta}^*) = \prod_{i=1}^{n} \left[\frac{\exp(\boldsymbol{z}_i^\top \boldsymbol{\beta}^*)}{\sum_{j \in R(t_i)} \exp(\boldsymbol{z}_j^\top \boldsymbol{\beta}^*)} \right]^{\delta_i}$, with $R(t_i) = \{k : t_k \geq t_i\}$. The corresponding log-partial likelihood is $\ell(\boldsymbol{\beta}^*) = \sum_{i=1}^{n} \delta_i \left[\boldsymbol{z}_i^\top \boldsymbol{\beta}^* - \log \sum_{j \in R(t_i)} \exp(\boldsymbol{z}_j^\top \boldsymbol{\beta}^*) \right]$.

Shrinkage priors

We explore several global–local shrinkage priors for $\boldsymbol{\beta}^*$.

Horseshoe (HS) [22, 23]:

$$\beta_j^* \mid \tau, \lambda_j \sim N(0, \tau^2 \lambda_j^2), \quad \lambda_j \sim \text{Half-Cauchy}(0, 1), \quad j = 1, \ldots, p, \tag{2}$$

where $\tau > 0$ controls global shrinkage, which can be fixed or assigned a half-Cauchy prior, and λ_j are local scales.

Regularized horseshoe (RHS) [24] adds a slab variance c^2:

$$\beta_j^* \mid \lambda_j, \tau, c \sim N(0, \tau^2 \tilde{\lambda}_j^2), \quad \tilde{\lambda}_j^2 = \frac{c^2 \lambda_j^2}{c^2 + \tau^2 \lambda_j^2}, \quad j = 1, \ldots, p, \tag{3}$$

with $\lambda_j \sim \text{Half-Cauchy}(0, 1)$, $\tau \sim \text{Half-Cauchy}(0, \tilde{\tau}_0^2)$ and $c^2 \sim \text{Inv-Gamma}(\nu/2, \nu s^2/2)$. We set $\nu = 4$, $s^2 = 2$ following [24]. For Cox models, $\tilde{\tau}_0$ is chosen via

$$\tilde{\tau}_0 = \frac{p_0}{p - p_0} \cdot \frac{1}{\sqrt{n_\delta}}, \tag{4}$$

where p_0 is the prior guess for the number of nonzero coefficients and n_δ is the number of uncensored events. The detailed derivation is provided in Appendix A.

When taxa exhibit phylogenetic relatedness, local scales can be given a conditional autoregressive (CAR) prior on $\psi_j = \log \lambda_j$:

$$p(\boldsymbol{\psi}) \propto \exp\left\{ -\frac{1}{2} \sum_{i \sim j} w_{ij} (\psi_i - \psi_j)^2 \right\}, \tag{5}$$

where the notation $\sum_{i \sim j}$ indicates summation over all pairs of connected taxa in the phylogenetic similarity graph, and w_{ij} measures similarity from a centered similarity matrix based on phylogenetic distances [25, 26]. Therefore, the prior for the local shrinkage parameter $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \ldots, \lambda_p)^\top$ is proportional to the product of the half-Cauchy dis-

tribution and the distribution in (5).

Bayesian LASSO [27]: In the Cox setting, we consider

$$\beta_j^* \mid \tau_j \sim N(0, \tau_j^2), \quad \tau_j^2 \sim \text{Gamma}\left(1, \frac{\lambda^2}{2}\right), \quad j = 1, \ldots, p, \tag{6}$$

marginally β_j^* follows Laplace$(0, 1/\lambda)$; λ may be fixed or given $\lambda^2 \sim \text{Gamma}(a_0, b_0)$.

Spike-and-Slab LASSO (SSLASSO) [28]:

$$\begin{aligned} \beta_j^* \mid \gamma_j, s_j &\sim \text{Laplace}(0, s_j), \quad s_j = (1-\gamma_j)s_0 + \gamma_j s_1, \\ \gamma_j &\sim \text{Bernoulli}(\theta), \quad \theta \sim \text{Beta}(1,1), \quad j = 1, \ldots, p. \end{aligned} \tag{7}$$

Here, $s_0 < s_1$ represent the spike and slab scales, γ_j indicates inclusion, and θ denotes the prior inclusion probability. Marginalizing γ_j gives a two-component Laplace mixture:

$$\beta_j^* \mid \theta \sim (1-\theta)\,\text{Laplace}(0, s_0) + \theta\,\text{Laplace}(0, s_1). \tag{8}$$

Posterior Inference

Let $p\left(\sum_{j=1}^p \beta_j^*\right)$ denote the "soft-centers" specified by Equation (1). The joint posterior is proportional to $\mathcal{L}(\boldsymbol{\beta}^*) \cdot p\left(\sum_{j=1}^p \beta_j^*\right)$ times the prior for $\boldsymbol{\beta}^*$ and prior hyperparameters. Firstly, for HS, we have

$$p\left(\boldsymbol{\beta}^*, \tau, \boldsymbol{\lambda} \mid \mathcal{O}\right) \propto \mathcal{L}(\boldsymbol{\beta}^*) p\left(\sum_{j=1}^p \beta_j^*\right) \pi(\boldsymbol{\beta}^* \mid \tau, \boldsymbol{\lambda}) \pi(\tau) \pi(\boldsymbol{\lambda}),$$

where $\pi(\boldsymbol{\beta}^* \mid \tau, \boldsymbol{\lambda})$, $\pi(\tau)$ and $\pi(\boldsymbol{\lambda})$ are defined by Equation (2). Secondly, for RHS, we have

$$p(\boldsymbol{\beta}^*, \tau, \boldsymbol{\lambda}, c^2 \mid \mathcal{O}) \propto \mathcal{L}(\boldsymbol{\beta}^*) p\left(\sum_{j=1}^p \beta_j^*\right) \pi(\boldsymbol{\beta}^* \mid \tau, \boldsymbol{\lambda}, c^2) \pi(\tau) \pi(\boldsymbol{\lambda}) \pi(c^2),$$

where $\pi(\boldsymbol{\beta}^* \mid \tau, \boldsymbol{\lambda}, c^2)$, $\pi(\tau)$, $\pi(\boldsymbol{\lambda})$ and $\pi(c^2)$ are defined by Equation (3), (4) and (5). Furthermore, for LASSO, we have

$$p(\boldsymbol{\beta}^*, \boldsymbol{\tau}, \lambda \mid \mathcal{O}) \propto \mathcal{L}(\boldsymbol{\beta}^*) p\left(\sum_{j=1}^p \beta_j^*\right) \pi(\boldsymbol{\beta}^* \mid \boldsymbol{\tau}, \lambda) \pi(\boldsymbol{\tau} \mid \lambda) \pi(\lambda),$$

where $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_p)^\top$ and $\pi(\boldsymbol{\beta}^* \mid \boldsymbol{\tau}, \lambda)$, $\pi(\boldsymbol{\tau} \mid \lambda)$, $\pi(\lambda)$ are defined by Equation (6). For the case of SSLASSO, we have

$$p(\boldsymbol{\beta}^*, \boldsymbol{\gamma}, \theta \mid \mathcal{O}) \propto \mathcal{L}(\boldsymbol{\beta}^*) p\left(\sum_{j=1}^{p} \beta_j^*\right) \pi(\boldsymbol{\beta}^* \mid \boldsymbol{\gamma}) \pi(\boldsymbol{\gamma} \mid \theta) \pi(\theta),$$

where $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_p)^\top$ and $\pi(\boldsymbol{\beta}^* \mid \boldsymbol{\gamma})$, $\pi(\boldsymbol{\gamma} \mid \theta)$, $\pi(\theta)$ are defined by Equation (7).

Posterior inference is conducted using the No-U-Turn Sampler (NUTS), a variant of Hamiltonian Monte Carlo implemented in Stan [29], which adaptively tunes step sizes and trajectory lengths. Discrete indicators (e.g. γ_j) in SSLASSO are integrated out using the marginal prior (8) to enable HMC.

From posterior draws we estimate $\boldsymbol{\beta}^*$ by the mean, component-wise median, or the maximum a posteriori (MAP): $\hat{\boldsymbol{\beta}}^*_{\text{mean}} = E(\boldsymbol{\beta}^* | \mathcal{O})$, $\hat{\boldsymbol{\beta}}^*_{\text{median}} = \left(\text{median}(\beta_j^* \mid \mathcal{O})\right)_{j=1}^{p}$, $\hat{\boldsymbol{\beta}}^*_{\text{MAP}} = \arg\max_{\boldsymbol{\beta}^*} p(\boldsymbol{\beta}^* | \mathcal{O})$, and construct 95% credible intervals from posterior quantiles.

3 Simulation Studies

3.1 Simulation Design

We evaluated the proposed method through simulations. We generated compositional covariates and the model coefficients adopted the same settings as Zhang et al. (2024) [4]. Time-to-event data are generated using the R package `simsurv` with $t \in [0, 100]$. Two baseline hazard functions were considered in the simulations: Weibull and Gompertz.

We considered different sample sizes $n \in \{200, 400\}$ and compositional covariate dimensions $p \in \{50, 150\}$. There are two different coefficient settings. The first setting has 6 non-zero compositional covariate coefficients, and the second setting has 12 non-zero coefficients. Under the first scenario, the 6 non-zero coefficients are β_j^* for $j = 16 + 2a$, $a = 1, 2, \ldots, 6$, with corresponding values 2.08, 1.31, -1.39, -1.86, 2.12, -2.26. Under the second scenario, the 12 non-zero coefficients are β_j^* for $j = 16 + 2a$, $a = 1, 2, \ldots, 12$, with corresponding values 2.08, -1.41, -1.39, -1.15, 2.12, 2.51, 1.31, -1.95, -1.86, 1.93, -1.34, -0.85 [4].

To obtain the compositional covariates, an $n \times p$ matrix $U = (u_{ij})$ was generated

from a multivariate normal distribution $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_p)$ and $\boldsymbol{\Sigma} = (\Sigma_{ij})$. The operational taxonomic unit (OTU) relative abundance matrix $X = (x_{ij})$ was obtained via the transformation $x_{ij} = (e^{u_{ij}}) \,/\, (\sum_{k=1}^{p} e^{u_{ik}})$. When the true coefficient is non-zero, we set $\mu_j = \log(0.5p)$; otherwise, $\mu_j = 0$. The covariance is defined as $\Sigma_{ij} = 0.75 - 0.015|i - j|$, so that the correlation between two covariates decreases with their distance. Additionally, the Bray-Curtis distance [30] was applied to calculate the dissimilarity matrix among the predictor variables.

The performance of the model is evaluated by calculating the mean squared error (MSE) between the estimated and the true $\boldsymbol{\beta}^*$. The posterior quantiles can be used to identify important taxonomic units. To assess the variable selection performance of the model, using the 95% credible intervals, we calculated the number of false positives (FP, the count of zero coefficients incorrectly identified as non-zero) and false negatives (FN, the count of non-zero coefficients incorrectly identified as zero). The Watanabe-Akaike Information Criterion (WAIC), Logarithm of the Pseudo-Marginal Likelihood (LPML) and the concordance Index (C-index) are calculated to demonstrate the quality of the model fit and prediction. WAIC measures the generalization ability of a model. A smaller WAIC value indicates a stronger predictive performance on unseen data, meaning the model is better. LPML measures a model's predictive performance on observed data. A larger LPML value indicates a better fit to the data. Both WAIC and LPML are computed using all posterior samples and do not rely on point estimates of the parameters. Moreover, the C-index represents the probability that the predicted risk scores are concordant with the actual event times. The C-index closer to 1 indicate better predictive performance.

Under the "soft-centers" constraint in Equation (1), we compared the performance of different priors, including the HS and RHS prior, the LASSO prior, and the SSLASSO prior. Several experimental setups were designed as follows:

1. **HS**. Assign a horseshoe prior to $\boldsymbol{\beta}^*$ as in Equation (2), with $\tau \sim \text{Half-Cauchy}(0, 1)$.

2. **RHS**. Assign a regularized horseshoe prior to $\boldsymbol{\beta}^*$ as in Equation (3). Calculate $\tilde{\tau}_0$ using $\tilde{\tau}_0 = \frac{p_0}{p - p_0} \cdot \frac{\tilde{\sigma}}{\sqrt{n}}$, and set $\tilde{\sigma}^2 \sim t(3, 0, 10)$ [24, 4].

3. **New-RHS**. Assign a regularized horseshoe prior to $\boldsymbol{\beta}^*$ as in Equation (3). Calculate $\tilde{\tau}_0$ using $\tilde{\tau}_0 = \frac{p_0}{p - p_0} \cdot \frac{\tilde{\sigma}}{\sqrt{n}}$, but set $\tilde{\sigma}^2 = 1$ directly.

4. **Est-RHS**. Assign a regularized horseshoe prior to $\boldsymbol{\beta}^*$ as in Equation (3). Calculate $\tilde{\tau}_0$ using $\tilde{\tau}_0 = \frac{p_0}{p-p_0} \cdot \frac{\tilde{\sigma}}{\sqrt{n}}$, set $\tilde{\sigma}^2 = 1/\mu$, and estimate μ by $\hat{\mu} = [\sum_{i=1}^{n} \exp(\boldsymbol{z}_i^\top \hat{\boldsymbol{\beta}})]/n$ [24].

5. **Cox-HS**. Assign a horseshoe prior to $\boldsymbol{\beta}^*$ as in Equation (2), but calculate $\tilde{\tau}_0$ using Equation (4) and set $\tau \sim$ Half-Cauchy$(0, \tilde{\tau}_0^2)$.

6. **Cox-RHS**. Assign a regularized horseshoe prior to $\boldsymbol{\beta}^*$ as in Equation (3), and calculate $\tilde{\tau}_0$ using Equation (4).

7. **LASSO**. Assign a prior to $\boldsymbol{\beta}^*$ as in Equation (6), with $\lambda^2 \sim$ Gamma$(1, 0.1)$ which makes its prior highly dispersed.

8. **SSLASSO**. Assign a prior to $\boldsymbol{\beta}^*$ as Equation (7), with $s_0 = 0.04$ and $s_1 = 0.5$, following Tang (2017) [31]. While not necessarily optimal, the choice of s_0 and s_1 ensures adequate shrinkage of zero coefficients while maintaining the values of the non-zero coefficients.

Additionally, we take $\nu = 4$ and $s^2 = 2$ for the prior of c^2 across all RHS priors. In each of the aforementioned settings, four chains were fitted, with the initial 2,000 iterations discarded as a warm-up phase and the subsequent 5,000 iterations retained for analysis. Moreover, we replicate the simulation 50 times and evaluate performance using the mean and standard deviation of MSE, FP, FN, WAIC, LPML and C-index.

3.2 Simulation Results

Partial simulation results for are presented in Table 1, Table 2, Table 3 and Table 4. The results are shown for the Weibull baseline function, where the posterior mean serves as the point estimate. Results for other settings are provided in the Appendix B. The results in the table represent the averages over 50 replicates, with the standard deviations given in parentheses.

To ensure the validity of posterior samples, graphical and numerical diagnostics were conducted on the convergence of the algorithm. The trace plots of posterior sample values against iterations, density plots, and autocorrelation function (ACF) plots were plotted. We also calculated the potential scale reduction factor (PSRF) $\hat{R}$ for each

Table 1. Simulation results for 6 non-zero coefficients setup under the Weibull baseline function (50 replicates). The point estimate is evaluated by the posterior mean.

n	p	**Prior**	**MSE**	**FP**	**FN**
200	50	HS	0.009(0.004)	0.800(0.990)	0.000(0.000)
		RHS	0.009(0.004)	0.740(0.965)	0.000(0.000)
		New-RHS	0.009(0.004)	0.640(0.875)	0.000(0.000)
		Est-RHS	0.011(0.005)	0.580(0.785)	0.000(0.000)
		Cox-HS	0.009(0.004)	0.800(0.969)	0.000(0.000)
		Cox-RHS	0.009(0.004)	0.760(0.960)	0.000(0.000)
		LASSO	0.010(0.004)	0.940(1.132)	0.000(0.000)
		SSLASSO	0.005(0.003)	0.100(0.303)	0.000(0.000)
	150	HS	0.007(0.004)	1.060(1.150)	0.000(0.000)
		RHS	0.007(0.003)	0.920(1.047)	0.000(0.000)
		New-RHS	0.006(0.003)	0.840(0.997)	0.000(0.000)
		Est-RHS	0.007(0.002)	0.700(0.863)	0.000(0.000)
		Cox-HS	0.007(0.003)	0.980(1.097)	0.000(0.000)
		Cox-RHS	0.007(0.002)	0.860(1.050)	0.000(0.000)
		LASSO	0.010(0.004)	1.980(1.790)	0.000(0.000)
		SSLASSO	0.011(0.007)	1.000(1.309)	0.000(0.000)
400	50	HS	0.004(0.002)	1.200(1.050)	0.000(0.000)
		RHS	0.004(0.002)	1.080(0.966)	0.000(0.000)
		New-RHS	0.004(0.002)	0.860(0.833)	0.000(0.000)
		Est-RHS	0.004(0.002)	1.000(0.948)	0.000(0.000)
		Cox-HS	0.004(0.002)	1.080(0.922)	0.000(0.000)
		Cox-RHS	0.004(0.002)	1.040(1.009)	0.000(0.000)
		LASSO	0.004(0.002)	1.360(1.064)	0.000(0.000)
		SSLASSO	0.003(0.002)	0.300(0.544)	0.000(0.000)
	150	HS	0.002(0.001)	1.580(1.386)	0.000(0.000)
		RHS	0.002(0.001)	1.400(1.400)	0.000(0.000)
		New-RHS	0.002(0.001)	1.420(1.430)	0.000(0.000)
		Est-RHS	0.002(0.001)	1.260(1.337)	0.000(0.000)
		Cox-HS	0.002(0.001)	1.480(1.297)	0.000(0.000)
		Cox-RHS	0.002(0.001)	1.300(1.298)	0.000(0.000)
		LASSO	0.003(0.001)	3.260(1.998)	0.000(0.000)
		SSLASSO	0.004(0.002)	1.400(1.400)	0.000(0.000)

Table 2. Simulation results for 12 non-zero coefficients setup under the Weibull baseline function (50 replicates). The point estimate is evaluated by the posterior mean.

n	p	Prior	MSE	FP	FN
200	50	HS	0.017(0.007)	1.260(1.175)	0.000(0.000)
		RHS	0.018(0.008)	1.180(1.240)	0.000(0.000)
		New-RHS	0.017(0.008)	1.140(1.212)	0.000(0.000)
		Est-RHS	0.022(0.010)	1.020(1.020)	0.000(0.000)
		Cox-HS	0.017(0.008)	1.240(1.222)	0.000(0.000)
		Cox-RHS	0.018(0.008)	1.240(1.135)	0.000(0.000)
		LASSO	0.016(0.007)	1.080(1.047)	0.000(0.000)
		SSLASSO	0.011(0.006)	0.060(0.314)	0.000(0.000)
	150	HS	0.022(0.006)	2.980(2.281)	0.100(0.303)
		RHS	0.026(0.009)	2.580(2.304)	0.100(0.303)
		New-RHS	0.022(0.006)	2.400(2.090)	0.100(0.303)
		Est-RHS	0.034(0.012)	1.840(1.633)	0.080(0.274)
		Cox-HS	0.023(0.007)	2.640(2.266)	0.100(0.303)
		Cox-RHS	0.026(0.009)	2.500(2.206)	0.100(0.303)
		LASSO	0.017(0.005)	2.740(2.239)	0.100(0.303)
		SSLASSO	0.010(0.006)	0.560(0.837)	0.020(0.141)
400	50	HS	0.006(0.002)	1.480(1.092)	0.000(0.000)
		RHS	0.006(0.002)	1.460(1.014)	0.000(0.000)
		New-RHS	0.006(0.002)	1.360(1.025)	0.000(0.000)
		Est-RHS	0.007(0.003)	1.340(1.062)	0.000(0.000)
		Cox-HS	0.006(0.003)	1.520(1.147)	0.000(0.000)
		Cox-RHS	0.006(0.002)	1.480(1.092)	0.000(0.000)
		LASSO	0.006(0.002)	1.180(0.962)	0.000(0.000)
		SSLASSO	0.004(0.002)	0.240(0.476)	0.000(0.000)
	150	HS	0.006(0.002)	3.980(2.190)	0.000(0.000)
		RHS	0.006(0.002)	3.680(2.142)	0.000(0.000)
		New-RHS	0.006(0.001)	3.340(1.944)	0.000(0.000)
		Est-RHS	0.006(0.002)	3.220(2.043)	0.000(0.000)
		Cox-HS	0.006(0.001)	3.660(2.153)	0.000(0.000)
		Cox-RHS	0.006(0.002)	3.560(2.130)	0.000(0.000)
		LASSO	0.005(0.001)	3.340(1.934)	0.000(0.000)
		SSLASSO	0.004(0.002)	1.060(1.202)	0.000(0.000)

Table 3. LPML, WAIC and C-index for 6 non-zero coefficients setup under Weibull baseline function (50 replicates). The point estimate is evaluated by the posterior mean.

n	p	Prior	LPML	WAIC	C-index
200	50	HS	701.410(18.184)	1400.335(36.514)	0.877(0.013)
		RHS	701.840(18.180)	1401.299(36.504)	0.877(0.013)
		New-RHS	700.759(17.968)	1399.272(36.095)	0.877(0.013)
		Est-RHS	702.625(18.027)	1402.973(36.187)	0.877(0.013)
		Cox-HS	701.264(18.198)	1400.170(36.538)	0.877(0.013)
		Cox-RHS	702.045(18.212)	1401.711(36.572)	0.877(0.013)
		LASSO	703.463(18.293)	1404.206(36.727)	0.878(0.013)
		SSLASSO	690.608(17.587)	1380.304(35.211)	0.873(0.013)
	150	HS	723.925(15.816)	1427.860(33.570)	0.910(0.010)
		RHS	724.515(15.840)	1431.079(33.390)	0.909(0.010)
		New-RHS	722.762(15.459)	1428.433(32.646)	0.908(0.010)
		Est-RHS	725.995(15.752)	1435.752(32.850)	0.908(0.010)
		Cox-HS	723.391(15.984)	1428.278(33.859)	0.909(0.010)
		Cox-RHS	724.670(16.033)	1431.426(33.796)	0.909(0.009)
		LASSO	738.296(16.622)	1445.310(36.213)	0.916(0.009)
		SSLASSO	710.139(15.498)	1405.872(31.581)	0.903(0.009)
400	50	HS	1640.735(21.886)	3280.674(43.796)	0.869(0.007)
		RHS	1640.788(21.891)	3280.800(43.804)	0.869(0.007)
		New-RHS	1640.035(21.858)	3279.324(43.734)	0.869(0.007)
		Est-RHS	1641.261(21.839)	3281.760(43.699)	0.869(0.007)
		Cox-HS	1640.274(21.871)	3279.780(43.763)	0.869(0.007)
		Cox-RHS	1640.596(21.900)	3280.417(43.821)	0.869(0.007)
		LASSO	1642.413(21.871)	3283.956(43.765)	0.869(0.007)
		SSLASSO	1630.794(21.940)	3261.231(43.888)	0.868(0.007)
	150	HS	1657.620(24.468)	3309.472(49.258)	0.885(0.007)
		RHS	1657.466(24.476)	3309.448(49.264)	0.885(0.007)
		New-RHS	1638.385(134.449)	3271.132(271.091)	0.885(0.008)
		Est-RHS	1658.207(24.383)	3311.120(49.050)	0.885(0.007)
		Cox-HS	1657.290(24.151)	3309.066(48.609)	0.885(0.007)
		Cox-RHS	1657.740(24.237)	3310.016(48.764)	0.885(0.007)
		LASSO	1675.339(24.916)	3340.781(50.241)	0.888(0.007)
		SSLASSO	1646.948(24.543)	3289.565(49.266)	0.883(0.007)

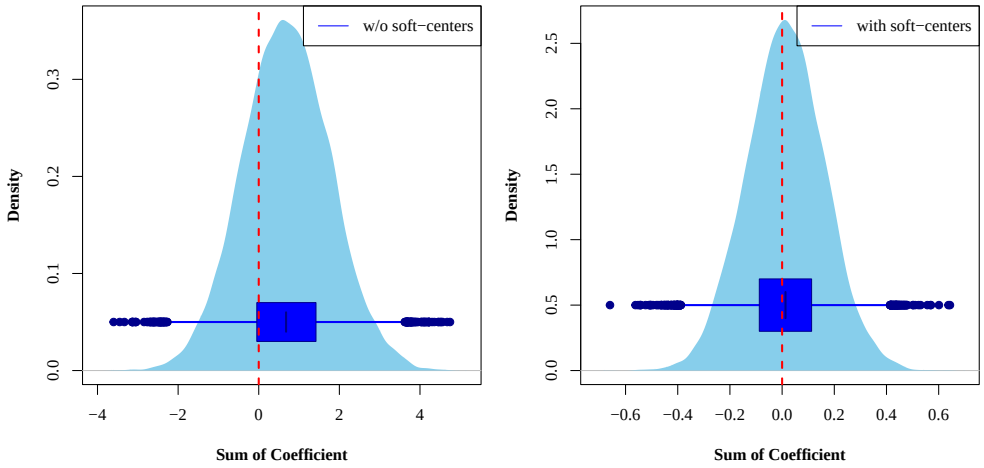
Table 4. LPML, WAIC and C-index for 12 non-zero coefficients setup under Weibull baseline function (50 replicates). The point estimate is evaluated by the posterior mean.

n	p	Prior	LPML	WAIC	C-index
200	50	HS	649.033(17.568)	1294.113(35.295)	0.905(0.007)
		RHS	649.999(17.834)	1296.083(35.857)	0.906(0.008)
		New-RHS	648.878(18.252)	1293.965(36.685)	0.906(0.008)
		Est-RHS	651.541(17.441)	1299.324(35.028)	0.905(0.007)
		Cox-HS	649.201(17.660)	1294.497(35.489)	0.905(0.007)
		Cox-RHS	649.978(17.513)	1296.027(35.185)	0.905(0.007)
		LASSO	646.745(17.554)	1289.945(35.270)	0.905(0.007)
		SSLASSO	634.481(17.224)	1267.753(34.493)	0.902(0.008)
	150	HS	708.275(18.216)	1365.637(41.711)	0.939(0.009)
		RHS	712.723(18.335)	1378.502(41.729)	0.937(0.009)
		New-RHS	707.422(17.866)	1369.523(40.145)	0.937(0.008)
		Est-RHS	718.815(18.402)	1397.242(40.907)	0.934(0.009)
		Cox-HS	708.380(17.678)	1368.243(40.767)	0.938(0.009)
		Cox-RHS	712.131(17.759)	1377.431(40.773)	0.937(0.009)
		LASSO	694.310(16.312)	1342.787(36.763)	0.939(0.008)
		SSLASSO	654.248(14.747)	1292.813(30.322)	0.926(0.008)
400	50	HS	1521.780(25.669)	3042.401(51.368)	0.898(0.006)
		RHS	1524.581(23.178)	3048.006(46.386)	0.897(0.006)
		New-RHS	1524.228(23.165)	3047.318(46.364)	0.897(0.006)
		Est-RHS	1523.015(25.604)	3044.896(51.235)	0.898(0.006)
		Cox-HS	1523.026(25.986)	3044.904(52.001)	0.898(0.006)
		Cox-RHS	1523.241(25.925)	3045.326(51.877)	0.898(0.006)
		LASSO	1521.223(25.949)	3041.377(51.923)	0.898(0.006)
		SSLASSO	1511.115(23.027)	3021.777(46.063)	0.896(0.006)
	150	HS	1574.351(28.002)	3133.726(56.596)	0.913(0.006)
		RHS	1575.509(27.936)	3136.301(56.490)	0.913(0.006)
		New-RHS	1573.203(27.513)	3132.206(55.659)	0.913(0.006)
		Est-RHS	1577.455(27.884)	3140.679(56.311)	0.913(0.006)
		Cox-HS	1574.640(27.245)	3134.562(55.065)	0.913(0.006)
		Cox-RHS	1575.802(27.281)	3136.932(55.137)	0.913(0.006)
		LASSO	1567.200(26.914)	3120.957(54.342)	0.913(0.006)
		SSLASSO	1529.080(26.349)	3053.321(52.907)	0.908(0.006)

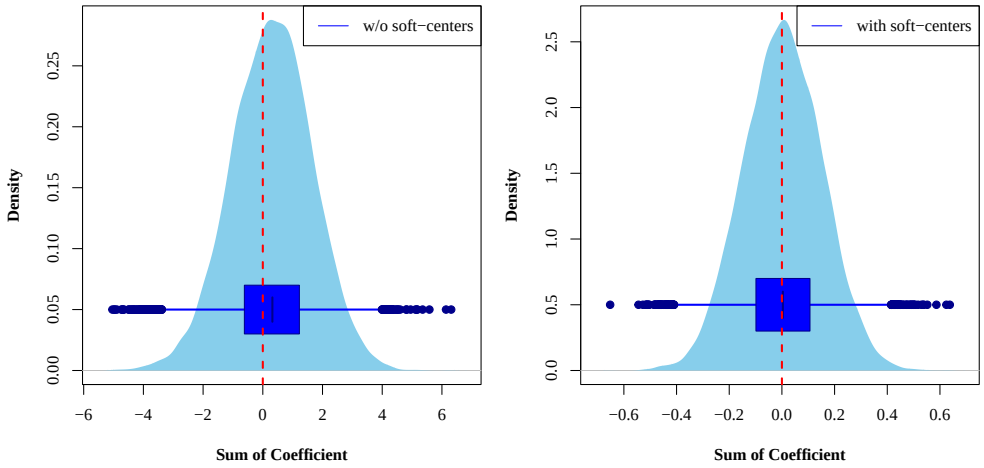
coefficient. The results are shown in Figure 5, Figure 6, Figure 7, Figure 8 and Figure 9 in Appendix B.

The test results show that when the sample size is sufficiently large, the sample paths exhibit stationarity with no obvious cycles or trends, the density plots show a normal distribution pattern, the autocorrelation coefficients decrease with the increase of iteration steps, and $\hat{R} < 1.1$ for each coefficient, close to 1, which indicates that the model has adequate convergence.

Moreover, to illustrate the role of "soft-centers", we also compared the results with and without "soft-centers" under the Cox-RHS prior. The evaluation was carried out for $n = 200, p = 150$ under the Weibull baseline function. Figure 1 shows the distribution of the sum of 150 coefficients across all 20,000 samples drawn from four chains. Regardless of whether there are 6 non-zero coefficients or 12, the results with "soft-centers" are closer to the requirement that the sum of coefficients be 0.



(a)



(b)

Figure 1: Distribution of the sum of 150 coefficients across all 20,000 samples drawn from four chains. The red dashed line indicates the value zero. Panels (a) and (b) correspond to scenarios with 6 and 12 non-zero coefficients, respectively. Results from the model without "soft-centers" are shown on the left in both figures.

4 Real Data Analysis

We analyzed the gut microbiome data from a nonobese diabetic (NOD) mice study [32, 17] to investigate the association between microbial taxa and the time to type 1 diabetes (T1D) onset. To reduce sparsity, we focused on the genus level and retained only taxa present in more than 5% of samples. The resulting dataset comprised metagenomic profiles of 173 mice across 30 bacterial taxa, including 55 that remained T1D-free and 118 that developed T1D. In addition to taxonomic abundances, the dataset included sex, antibiotic administration, and survival time to T1D onset. Since logarithmic transfor-

mation requires strictly positive values, we replaced zero abundances with half of the minimum observed abundance before computing relative compositions.

Hyperparameters of the priors were tuned by cross-validation using the C-index. For HS and RHS priors, the C-index was robust to different hyperparameter values, so we adopted the same settings as in the simulations. For the LASSO prior, we fixed $a_0 = 1$ and $b_0 = 0.1$, which provided adequate shrinkage and stable performance. For SSLASSO, we evaluated s_0 over two ranges, $\{0.04 \pm 0.01k : k = 0, 1, 2, 3\}$ and $\{1 \pm 0.1k : k = 0, 1, 2, 3\}$, while fixing $s_1 \in \{1, 10\}$. The final specification was $s_0 = 1$ and $s_1 = 10$.

We fitted the model using four chains, each with 4,000 warm-up iterations and 5,000 sampling iterations. Convergence diagnostics indicated satisfactory mixing, as reflected in the $\hat{R}$ values (Figure 3). Posterior means, medians, and MAP estimates were summarized as point estimates. The 95% credible intervals were constructed from the 2.5% and 97.5% quantiles. Figure 2 presents posterior means with credible intervals for all taxa, while results based on medians and MAP estimates are reported in Appendix C.

To evaluate predictive performance, we compared WAIC, LPML, and the C-index in Table 5 and Figure 4. In addition, we performed leave-one-out (LOO) cross-validation using the R package `loo` in Table 6, which provided the Expected Log pointwise Predictive Density (ELPD), its difference from the best-performing model (ELPD Diff), the effective number of parameters (PLOO), and leave-one-out Information Criterion (LOOIC).

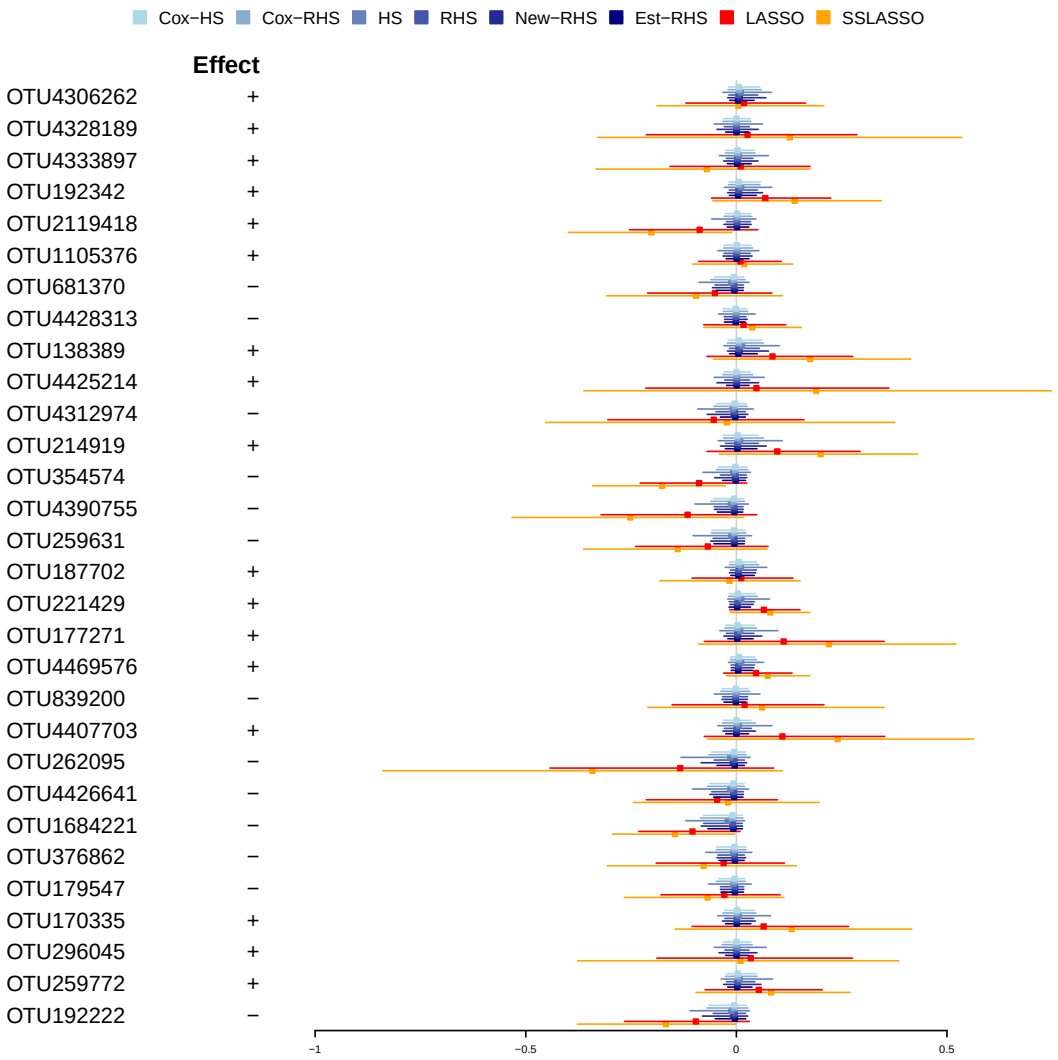


Figure 2: Coefficient estimated by posterior mean for each compositional variable.

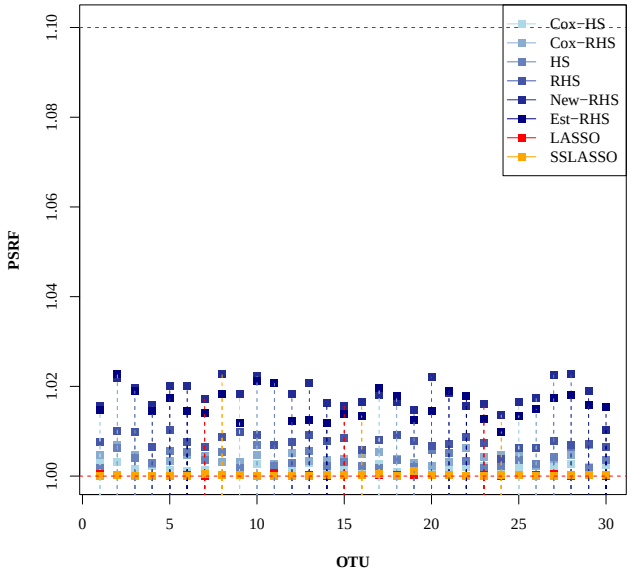


Figure 3: $\hat{R}$ for each model. The positions of 1 and 1.1 are marked with red dashed lines. $\hat{R}$ of all models are less than 1.1 and relatively close to 1, which confirms the convergence of the algorithm.

The results reveal a trade-off between model fit and predictive generalization. SSLASSO achieved the highest LPML and C-index, but its WAIC and LOOIC values were substantially larger, indicating potential overfitting due to excessive model complexity. In contrast, HS and Cox-RHS priors attained lower WAIC values and comparable predictive accuracy, suggesting a more balanced compromise between fit and parsimony. The LOOIC analysis further confirmed this. Cox-HS, Cox-RHS, HS, RHS, New-RHS, and Est-RHS exhibited similar predictive performance, whereas LASSO and SSLASSO displayed inflated PLOO values, again reflecting higher complexity.

Table 5. Bayesian evaluation metrics for models.

Prior	**LPML**	**WAIC**
HS	555.7138	1111.2764
RHS	556.0356	1112.0032
New-RHS	556.1391	1112.1770
Est-RHS	556.0102	1111.9686
Cox-HS	555.8637	1111.6522
Cox-RHS	555.8247	1111.5652
LASSO	557.5628	1113.9153
SSLASSO	563.6911	1123.9059

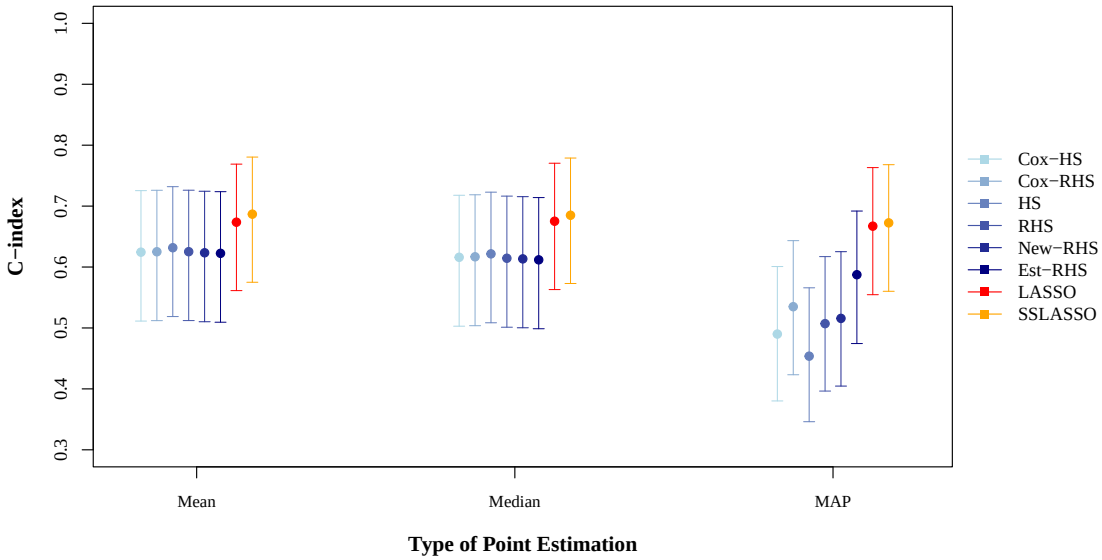


Figure 4: The C-index values of the model under different priors and different estimations.

Table 6. Results of leave-one-out cross-validation. The values of each metrics and their standard error are displayed.

Prior	ELPD Diff	ELPD	PLOO	LOOIC
HS	0.00(0.00)	-555.71(29.17)	4.86(0.34)	1111.43(58.34)
RHS	-0.32(0.81)	-556.04(29.16)	2.21(0.15)	1112.07(58.31)
New-RHS	-0.43(0.75)	-556.14(29.16)	2.66(0.19)	1112.28(58.33)
Est-RHS	-0.30(0.89)	-556.01(29.15)	1.87(0.13)	1112.02(58.31)
Cox-HS	-0.15(0.70)	-555.86(29.15)	2.38(0.16)	1111.73(58.30)
Cox-RHS	-0.11(0.61)	-555.83(29.15)	2.65(0.18)	1111.65(58.30)
LASSO	-1.85(3.22)	-557.57(29.54)	19.92(1.55)	1115.14(59.07)
SSLASSO	-7.99(5.85)	-563.70(30.27)	31.09(2.61)	1127.40(60.55)

In summary, these results highlight the potential role of specific gut microbial taxa in modulating the risk of T1D. As illustrated in Figure 2, the analysis revealed 17 OTUs exhibiting positive associations and 13 OTUs showing negative associations with the time to T1D onset. Notably, according to the SSLASSO, four negatively associated OTUs (OTU2119418, OTU354574, OTU1684221, and OTU192222) were identified as statistically significant, as their 95% credible intervals excluded zero. Moreover, in addition to the previously identified taxa, there are two additional negatively associated OTUs (OTU4390755 and OTU262095) that are significant at the 90% credible interval level.

5 Discussion

We propose a Bayesian Cox proportional hazards model for the analysis of microbiome compositional data. First, to address the sum-to-one constraint of the compositional covariates, we directly apply a logarithmic transformation to the original compositional covariates. The transformation converts the covariate constraint into a soft sum-to-zero condition on the coefficients, while preserving dimensional interpretability. Second, to address the variable selection problem in high dimensional sparse scenarios, we investigate several priors, including horseshoe prior and its regularization variant, Bayesian LASSO, and the Spike-and-Slab LASSO to the constrained coefficients. For the regularized horseshoe prior, we extend the concept of effective sample size from linear models to Cox models. This allows us to set hyperparameters more reasonably and enables the prior to adaptively adjust sparsity according to the number of non-censored observations. Third, to address phylogenetic relationships in microbiome data, we embed a conditional autoregressive structure into the local shrinkage parameters of the horseshoe prior. This structure is leveraged during variable selection to better manage correlations among covariates. Results from both simulation studies and real data analyses confirm that the proposed Cox model achieves adequate practical value.

The model constructed in this paper currently only includes compositional covariates. However, in practical applications, data often contains both compositional covariates and non-compositional covariates (such as continuous variables like height and weight, and binary variables like gender). Incorporating these non-compositional variables into the model is expected to further improve prediction performance. A straightforward approach is to incorporate them into the model and assign weakly informative prior (e.g., Cauchy(0, 2.5) [4]). Moreover, current applications of the LASSO and Spike-and-Slab LASSO do not account for phylogenetic information. Representing the prior hierarchically and introducing a CAR structure into the local shrinkage parameters may provide a more effective way to capture phylogenetic dependencies in compositional data.

The method presented in this paper still has several limitations. Currently, many results related to the horseshoe prior and its variants have been studied primarily in the context of generalized linear models. But their behavior in non-linear models such as the Cox model requires further investigation. If more rigorous mathematical methods

could be employed to generalize the prior hyperparameters for global shrinkage parameters from linear model to Cox model, it may enhance the horseshoe prior's adaptive capability regarding model sparsity.

Another limitation is that this paper performs parameter estimation for the Cox model within a partial likelihood framework, avoiding the need to model the baseline hazard function. If a full likelihood approach were adopted, how to use existing methods, such as the Spline-Based method, or select an appropriate prior for the baseline hazard function to further enhance model performance remains an open question requiring further exploration. Additionally, future research could explore other types of high dimensional sparse priors. Different priors may be suitable for different fields and application scenarios. Researchers can employ a combination of model selection metrics to choose the most appropriate prior for a specific problem.

Our approach offers a potentially valuable tool for survival analysis in microbiome research, particularly for identifying taxa associated with time-to-event outcomes.

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Appendix

A Derivation of $\tilde{\tau}_0$ for the Cox Model

Let the observed survival data be $\mathcal{O} = \{(t_i, \delta_i, \boldsymbol{x}_i) : i = 1, 2, \ldots, n\}$, where t_i is the observed event or censoring time, δ_i is the event indicator, and $\boldsymbol{x}_i$ denotes the compositional covariates. Assume the horseshoe prior for the coefficient vector $\boldsymbol{\beta}^*$ is specified as $\boldsymbol{\beta}^* \mid \boldsymbol{\lambda}, \tau \sim N(\mathbf{0}, \tau^2 \boldsymbol{\Lambda})$, with $\boldsymbol{\Lambda} = \text{diag}(\lambda_1^2, \ldots, \lambda_p^2)$, where $\tau > 0$ is the global shrinkage parameter and $\lambda_j > 0$ are the local shrinkage parameters.

Step 1: Partial likelihood derivatives

Define, for $k = 0, 1, 2$, $\mathcal{S}^{(k)}(\boldsymbol{\beta}^*, t) = \sum_{j \in R(t)} \boldsymbol{z}_j^{\otimes k} \exp\{\boldsymbol{z}_j^\top \boldsymbol{\beta}^*\}$, where $\boldsymbol{z}_j^{\otimes 0} = 1$, $\boldsymbol{z}_j^{\otimes 1} = \boldsymbol{z}_j$ and $\boldsymbol{z}_j^{\otimes 2} = \boldsymbol{z}_j \boldsymbol{z}_j^\top$. Let $\overline{\boldsymbol{z}}(\boldsymbol{\beta}^*, t) = \frac{\mathcal{S}^{(1)}(\boldsymbol{\beta}^*, t)}{\mathcal{S}^{(0)}(\boldsymbol{\beta}^*, t)}$, $V(\boldsymbol{\beta}^*, t) = \frac{\mathcal{S}^{(2)}(\boldsymbol{\beta}^*, t)}{\mathcal{S}^{(0)}(\boldsymbol{\beta}^*, t)} - \overline{\boldsymbol{z}}(\boldsymbol{\beta}^*, t)\, \overline{\boldsymbol{z}}(\boldsymbol{\beta}^*, t)^\top$, so that $\overline{\boldsymbol{z}}(\cdot)$ and $V(\cdot)$ represent the weighted covariate mean and covariance matrix for the risk set at time t.

The score vector and Hessian matrix of the Cox partial log-likelihood are $U(\boldsymbol{\beta}^*) = \sum_{i=1}^n \delta_i \left(\boldsymbol{z}_i - \overline{\boldsymbol{z}}(\boldsymbol{\beta}^*, t_i)\right)$, $H(\boldsymbol{\beta}^*) = -\sum_{i=1}^n \delta_i V(\boldsymbol{\beta}^*, t_i)$.

Step 2: Quadratic approximation around $\hat{\boldsymbol{\beta}}$

Let $\hat{\boldsymbol{\beta}}$ denote the partial maximum likelihood estimate (PMLE), i.e., $U(\hat{\boldsymbol{\beta}}) = 0$. Denote the observed information matrix $\mathcal{I}(\hat{\boldsymbol{\beta}}) = -H(\hat{\boldsymbol{\beta}}) = \sum_{i=1}^n \delta_i V(\hat{\boldsymbol{\beta}}, t_i)$. Using a second-order Taylor expansion of $\ell(\boldsymbol{\beta}^*)$ about $\hat{\boldsymbol{\beta}}$, we have $\ell(\boldsymbol{\beta}^*) \approx \ell(\hat{\boldsymbol{\beta}}) - \frac{1}{2}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}})^\top \mathcal{I}(\hat{\boldsymbol{\beta}})(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}) + \text{const.}$

Step 3: Posterior distribution with the HS prior

Combining the Gaussian approximation to the partial likelihood with the horseshoe prior, the unnormalized posterior density (conditional on $\boldsymbol{\Lambda}$, τ, and $\mathcal{O}$) is

$$p(\boldsymbol{\beta}^* \mid \boldsymbol{\lambda}, \tau, \mathcal{O}) \propto \exp\left\{-\tfrac{1}{2}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}})^\top \mathcal{I}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}) - \tfrac{1}{2\tau^2} \boldsymbol{\beta}^{*\top} \boldsymbol{\Lambda}^{-1} \boldsymbol{\beta}^*\right\} \propto N\left(\overline{\boldsymbol{\beta}},\ \Sigma\right),$$

with posterior mean and covariance $\overline{\boldsymbol{\beta}} = \left(\mathcal{I} + \tau^{-2}\boldsymbol{\Lambda}^{-1}\right)^{-1} \mathcal{I}\, \hat{\boldsymbol{\beta}}$, $\quad \Sigma = \left(\mathcal{I} + \tau^{-2}\boldsymbol{\Lambda}^{-1}\right)^{-1}$. This may also be written as $\overline{\boldsymbol{\beta}} = \left[I_p + \tau^{-2}\mathcal{I}^{-1}\boldsymbol{\Lambda}^{-1}\right]^{-1} \hat{\boldsymbol{\beta}}$, where I_p is the $p \times p$ identity matrix.

Step 4: Diagonal approximation

When covariates are assumed uncorrelated, $\mathcal{I}$ can be approximated by a diagonal matrix. Denote $\mathcal{I}_{dd} = \sum_{i=1}^n \delta_i \left[\frac{\sum_{j \in R(t_i)} z_{jd}^2 \exp(\boldsymbol{z}_j^\top \hat{\boldsymbol{\beta}})}{\sum_{j \in R(t_i)} \exp(\boldsymbol{z}_j^\top \hat{\boldsymbol{\beta}})} - \left(\frac{\sum_{j \in R(t_i)} z_{jd} \exp(\boldsymbol{z}_j^\top \hat{\boldsymbol{\beta}})}{\sum_{j \in R(t_i)} \exp(\boldsymbol{z}_j^\top \hat{\boldsymbol{\beta}})}\right)^2\right]$, $d =$

$1, \ldots, p$, then $\mathcal{I} \approx \text{diag}(\mathcal{I}_{11}, \ldots, \mathcal{I}_{pp})$.

Under this approximation, the d-th component of the posterior mean becomes $\overline{\beta}_d \approx \frac{1}{1+\tau^{-2}\lambda_d^{-2}\mathcal{I}_{dd}^{-1}}\,\hat{\beta}_d = (1-\kappa_d)\,\hat{\beta}_d$, where the shrinkage factor is $\kappa_d = \frac{1}{1+\tau^2\lambda_d^2\mathcal{I}_{dd}}$.

Step 5: Distribution of the shrinkage factor

Defining $a_d^2 = \tau^2\,\mathcal{I}_{dd}$, the shrinkage factor can be written as $\kappa_d = \{1 + a_d^2\lambda_d^2\}^{-1}$. When $\lambda_d \sim$ Half-Cauchy$(0, 1)$, the induced prior for κ_d (Piironen & Vehtari, 2017) is $p(\kappa_d \mid \tau) = \frac{1}{\pi}\frac{a_d}{(a_d^2-1)\kappa_d+1}\frac{1}{\sqrt{\kappa_d}\sqrt{1-\kappa_d}}$, with mean and variance $E(\kappa_d \mid \tau) = \frac{1}{1+a_d}$, $\text{Var}(\kappa_d \mid \tau) = \frac{a_d}{2(1+a_d)^2}$.

Step 6: Effective number of nonzero coefficients

Following Piironen & Vehtari (2017), the effective sample size is $m_{\text{eff}} = \sum_{d=1}^{p}(1-\kappa_d)$, which yields $E(m_{\text{eff}} \mid \tau) = \sum_{d=1}^{p}\frac{a_d}{1+a_d}$, $\text{Var}(m_{\text{eff}} \mid \tau) = \sum_{d=1}^{p}\frac{a_d}{2(1+a_d)^2}$.

Step 7: Simplified case $V_d(\hat{\boldsymbol{\beta}}, t_i) \equiv 1$

If we further assume $V_d(\hat{\boldsymbol{\beta}}, t_i) = 1$ for all d, then $\mathcal{I}_{dd} \approx n_\delta$, where $n_\delta = \sum_{i=1}^{n}\delta_i$ is the number of uncensored events. In this case, $E(m_{\text{eff}} \mid \tau) = \frac{p\,\tau\sqrt{n_\delta}}{1+\tau\sqrt{n_\delta}}$, $\text{Var}(m_{\text{eff}} \mid \tau) = \frac{p\,\tau\sqrt{n_\delta}}{2(1+\tau\sqrt{n_\delta})^2}$.

Setting $E(m_{\text{eff}} \mid \tau) = p_0$, where p_0 is the prior guess for the number of relevant predictors, and solving for τ yields the expression in Equation (4): $\tilde{\tau}_0 = \frac{p_0}{p-p_0}\cdot\frac{1}{\sqrt{n_\delta}}$.

B Supplementary Results of Simulation Studies

B.1 Results for Gompertz Baseline Function

Table 7. Simulation results for 6 non-zero coefficients setup under the Gompertz baseline function (50 replicates). The point estimate is evaluated by the posterior mean.

n	p	Prior	MSE	FP	FN
200	50	HS	0.009(0.004)	0.900(1.165)	0.000(0.000)
		RHS	0.009(0.004)	0.760(1.001)	0.000(0.000)
		New-RHS	0.009(0.004)	0.660(0.823)	0.000(0.000)
		Est-RHS	0.010(0.005)	0.640(0.875)	0.000(0.000)
		Cox-HS	0.009(0.004)	0.780(0.996)	0.000(0.000)
		Cox-RHS	0.009(0.004)	0.780(1.016)	0.000(0.000)
		LASSO	0.010(0.004)	1.060(1.219)	0.000(0.000)
		SSLASSO	0.005(0.003)	0.120(0.328)	0.000(0.000)
	150	HS	0.007(0.004)	1.060(1.132)	0.000(0.000)
		RHS	0.007(0.003)	0.900(1.035)	0.000(0.000)
		New-RHS	0.006(0.003)	0.840(0.997)	0.000(0.000)
		Est-RHS	0.006(0.002)	0.640(0.942)	0.000(0.000)
		Cox-HS	0.007(0.004)	1.020(1.078)	0.000(0.000)
		Cox-RHS	0.007(0.003)	0.900(1.035)	0.000(0.000)
		LASSO	0.010(0.004)	2.100(1.764)	0.000(0.000)
		SSLASSO	0.011(0.007)	1.020(1.169)	0.000(0.000)
400	50	HS	0.004(0.002)	1.160(0.976)	0.000(0.000)
		RHS	0.004(0.002)	1.020(0.937)	0.000(0.000)
		New-RHS	0.004(0.002)	0.900(0.839)	0.000(0.000)
		Est-RHS	0.004(0.002)	0.860(0.904)	0.000(0.000)
		Cox-HS	0.004(0.002)	1.020(0.892)	0.000(0.000)
		Cox-RHS	0.004(0.002)	1.000(0.948)	0.000(0.000)
		LASSO	0.004(0.002)	1.360(1.005)	0.000(0.000)
		SSLASSO	0.003(0.002)	0.340(0.626)	0.000(0.000)
	150	HS	0.002(0.001)	1.540(1.515)	0.000(0.000)
		RHS	0.002(0.001)	1.320(1.406)	0.000(0.000)
		New-RHS	0.002(0.001)	1.240(1.379)	0.000(0.000)
		Est-RHS	0.002(0.001)	1.180(1.351)	0.000(0.000)
		Cox-HS	0.002(0.001)	1.360(1.382)	0.000(0.000)
		Cox-RHS	0.002(0.001)	1.280(1.325)	0.000(0.000)
		LASSO	0.003(0.001)	3.340(2.264)	0.000(0.000)
		SSLASSO	0.004(0.002)	1.300(1.298)	0.000(0.000)

Table 8. Simulation results for 12 non-zero coefficients setup under the Gompertz baseline function (50 replicates). The point estimate is evaluated by the posterior mean.

n	p	Prior	MSE	FP	FN
200	50	HS	0.016(0.007)	1.160(1.184)	0.000(0.000)
		RHS	0.017(0.008)	1.040(1.160)	0.000(0.000)
		New-RHS	0.017(0.008)	1.020(1.097)	0.000(0.000)
		Est-RHS	0.020(0.010)	1.000(1.069)	0.000(0.000)
		Cox-HS	0.016(0.007)	1.080(1.066)	0.000(0.000)
		Cox-RHS	0.017(0.008)	1.080(1.140)	0.000(0.000)
		LASSO	0.015(0.006)	0.920(0.986)	0.000(0.000)
		SSLASSO	0.011(0.006)	0.100(0.303)	0.000(0.000)
	150	HS	0.021(0.006)	3.220(2.418)	0.080(0.274)
		RHS	0.024(0.009)	2.600(2.010)	0.080(0.274)
		New-RHS	0.020(0.006)	2.440(1.831)	0.080(0.274)
		Est-RHS	0.031(0.012)	1.900(1.644)	0.080(0.274)
		Cox-HS	0.021(0.007)	2.780(2.073)	0.080(0.274)
		Cox-RHS	0.024(0.009)	2.540(2.043)	0.080(0.274)
		LASSO	0.016(0.005)	2.640(2.136)	0.100(0.303)
		SSLASSO	0.010(0.006)	0.580(0.835)	0.000(0.000)
400	50	HS	0.006(0.002)	1.700(1.015)	0.000(0.000)
		RHS	0.006(0.002)	1.600(0.904)	0.000(0.000)
		New-RHS	0.006(0.002)	1.460(0.930)	0.000(0.000)
		Est-RHS	0.006(0.003)	1.520(1.015)	0.000(0.000)
		Cox-HS	0.006(0.003)	1.480(0.863)	0.000(0.000)
		Cox-RHS	0.006(0.002)	1.660(0.961)	0.000(0.000)
		LASSO	0.006(0.003)	1.240(0.822)	0.000(0.000)
		SSLASSO	0.004(0.002)	0.300(0.544)	0.000(0.000)
	150	HS	0.005(0.001)	4.040(2.356)	0.000(0.000)
		RHS	0.006(0.001)	3.900(2.323)	0.000(0.000)
		New-RHS	0.005(0.001)	3.700(2.279)	0.000(0.000)
		Est-RHS	0.006(0.002)	3.480(2.225)	0.000(0.000)
		Cox-HS	0.005(0.001)	3.840(2.280)	0.000(0.000)
		Cox-RHS	0.005(0.001)	3.860(2.321)	0.000(0.000)
		LASSO	0.005(0.001)	3.620(2.320)	0.000(0.000)
		SSLASSO	0.004(0.002)	0.980(0.937)	0.000(0.000)

Table 9. LPML, WAIC and C-index for 6 non-zero coefficients setup under Gompertz baseline function (50 replicates). The point estimate is evaluated by the posterior mean.

n	p	Prior	LPML	WAIC	C-index
200	50	HS	709.091(16.040)	1415.694(32.226)	0.877(0.013)
		RHS	709.521(16.039)	1416.646(32.225)	0.877(0.013)
		New-RHS	708.459(15.854)	1414.660(31.865)	0.877(0.013)
		Est-RHS	710.269(15.870)	1418.248(31.880)	0.877(0.013)
		Cox-HS	708.734(16.037)	1415.091(32.221)	0.877(0.013)
		Cox-RHS	709.496(16.006)	1416.597(32.160)	0.877(0.013)
		LASSO	710.943(16.124)	1419.168(32.391)	0.877(0.013)
		SSLASSO	698.331(15.439)	1395.740(30.922)	0.873(0.013)
	150	HS	729.693(14.196)	1439.382(30.629)	0.910(0.010)
		RHS	730.313(14.233)	1442.568(30.414)	0.908(0.010)
		New-RHS	728.184(13.977)	1439.436(29.816)	0.908(0.010)
		Est-RHS	731.762(14.110)	1447.228(29.764)	0.907(0.010)
		Cox-HS	729.139(14.342)	1439.677(30.818)	0.909(0.010)
		Cox-RHS	730.396(14.356)	1442.781(30.717)	0.909(0.010)
		LASSO	744.054(15.065)	1456.774(33.345)	0.916(0.010)
		SSLASSO	715.882(13.967)	1417.430(28.601)	0.903(0.009)
400	50	HS	1661.927(18.170)	3323.047(36.360)	0.869(0.007)
		RHS	1661.956(18.156)	3323.124(36.333)	0.869(0.007)
		New-RHS	1661.227(18.165)	3321.698(36.350)	0.869(0.007)
		Est-RHS	1662.410(18.141)	3324.045(36.302)	0.869(0.007)
		Cox-HS	1661.790(18.146)	3322.798(36.313)	0.869(0.007)
		Cox-RHS	1662.102(18.222)	3323.415(36.464)	0.869(0.007)
		LASSO	1663.908(18.144)	3326.935(36.307)	0.869(0.007)
		SSLASSO	1651.995(18.189)	3303.625(36.389)	0.867(0.007)
	150	HS	1680.258(18.590)	3354.777(37.564)	0.884(0.007)
		RHS	1680.035(18.556)	3354.611(37.490)	0.884(0.007)
		New-RHS	1679.116(18.458)	3352.947(37.287)	0.884(0.007)
		Est-RHS	1680.789(18.464)	3356.321(37.263)	0.884(0.007)
		Cox-HS	1679.906(18.258)	3354.352(36.862)	0.884(0.007)
		Cox-RHS	1680.377(18.241)	3355.318(36.833)	0.884(0.007)
		LASSO	1698.305(19.204)	3386.764(38.857)	0.887(0.007)
		SSLASSO	1669.687(18.566)	3335.027(37.330)	0.882(0.007)

Table 10. LPML, WAIC and C-index for 12 non-zero coefficients setup under Gompertz baseline function (50 replicates). The point estimate is evaluated by the posterior mean.

n	p	Prior	LPML	WAIC	C-index
200	50	HS	672.612(12.958)	1341.395(26.050)	0.904(0.007)
		RHS	672.962(13.752)	1342.119(27.644)	0.904(0.008)
		New-RHS	672.375(13.471)	1341.064(27.097)	0.904(0.008)
		Est-RHS	675.046(12.803)	1346.427(25.726)	0.904(0.007)
		Cox-HS	672.820(12.987)	1341.842(26.113)	0.904(0.007)
		Cox-RHS	673.595(12.915)	1343.367(25.976)	0.904(0.007)
		LASSO	670.398(12.852)	1337.341(25.845)	0.904(0.007)
		SSLASSO	657.555(12.566)	1313.902(25.191)	0.901(0.008)
	150	HS	729.682(16.587)	1409.741(38.566)	0.937(0.008)
		RHS	734.011(16.720)	1422.000(38.601)	0.935(0.009)
		New-RHS	727.706(15.979)	1411.367(36.806)	0.936(0.008)
		Est-RHS	739.721(16.333)	1439.398(37.071)	0.933(0.009)
		Cox-HS	729.932(16.500)	1412.443(38.387)	0.936(0.008)
		Cox-RHS	733.669(16.498)	1421.385(38.232)	0.935(0.009)
		LASSO	716.124(15.019)	1387.749(34.393)	0.937(0.008)
		SSLASSO	675.425(13.058)	1335.735(26.935)	0.924(0.008)
400	50	HS	1580.049(18.422)	3158.952(36.870)	0.897(0.006)
		RHS	1583.155(17.700)	3165.165(35.424)	0.896(0.006)
		New-RHS	1582.844(17.730)	3164.563(35.483)	0.896(0.006)
		Est-RHS	1581.252(18.390)	3161.377(36.802)	0.897(0.006)
		Cox-HS	1580.677(18.863)	3160.214(37.748)	0.897(0.006)
		Cox-RHS	1580.920(18.738)	3160.698(37.500)	0.897(0.006)
		LASSO	1578.842(18.804)	3156.628(37.632)	0.897(0.006)
		SSLASSO	1567.276(18.297)	3134.086(36.603)	0.896(0.006)
	150	HS	1634.243(20.216)	3254.122(40.993)	0.911(0.006)
		RHS	1635.332(20.220)	3256.529(41.003)	0.911(0.006)
		New-RHS	1632.547(20.056)	3251.544(40.686)	0.911(0.006)
		Est-RHS	1637.189(20.038)	3260.670(40.606)	0.911(0.006)
		Cox-HS	1634.780(19.627)	3255.434(39.778)	0.911(0.006)
		Cox-RHS	1635.966(19.590)	3257.809(39.727)	0.911(0.006)
		LASSO	1627.495(19.146)	3242.062(38.809)	0.911(0.006)
		SSLASSO	1589.352(17.784)	3173.922(35.814)	0.906(0.006)

B.2 Algorithm convergence diagnostics in simulation studies

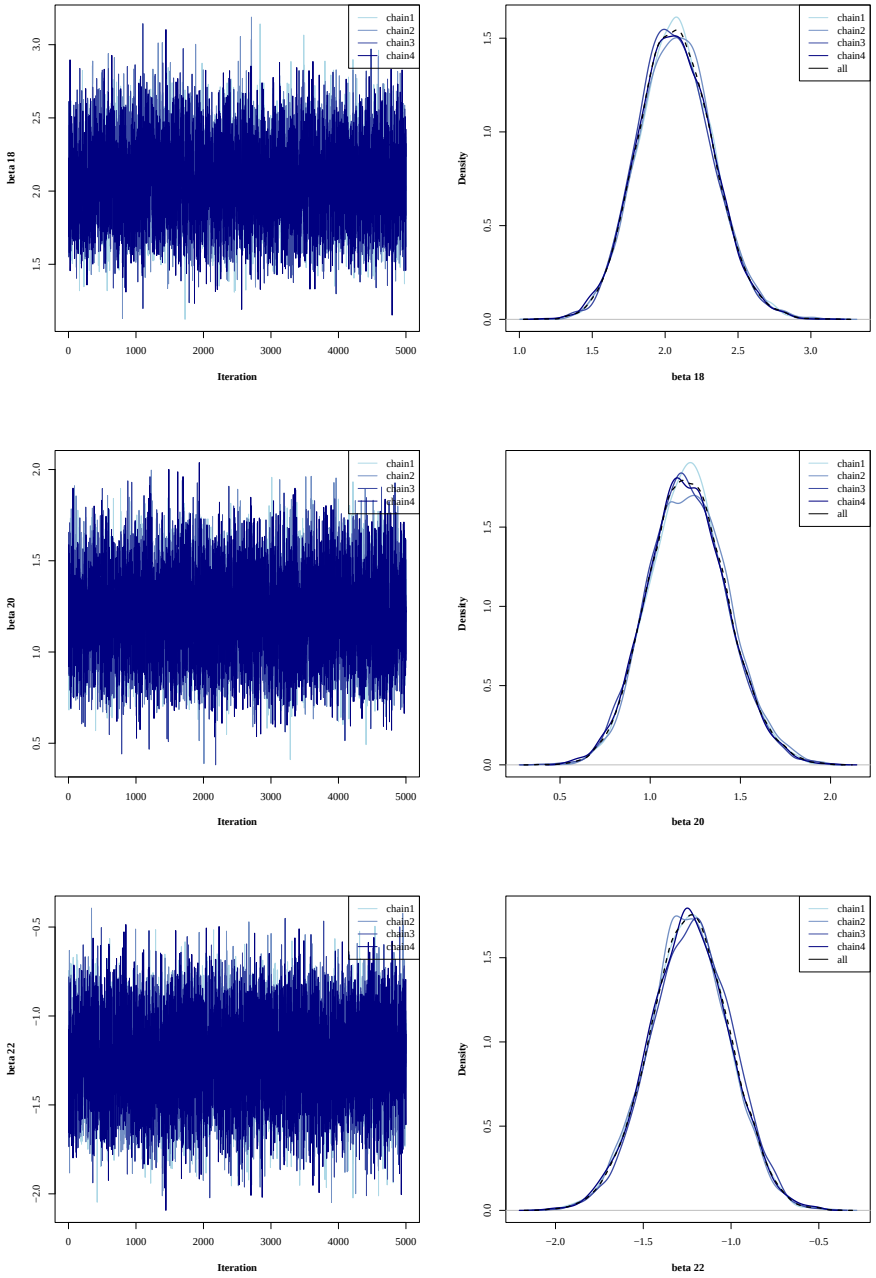


Figure 5: Trace plot (left) and density plot (right) for β^*_{18}, β^*_{20} and β^*_{22} in the case of 6 non-zero coefficients of the Cox-RHS prior with a Weibull baseline function while $n = 200$, $p = 150$. The black dashed line in the density plot represents the case of all samples.

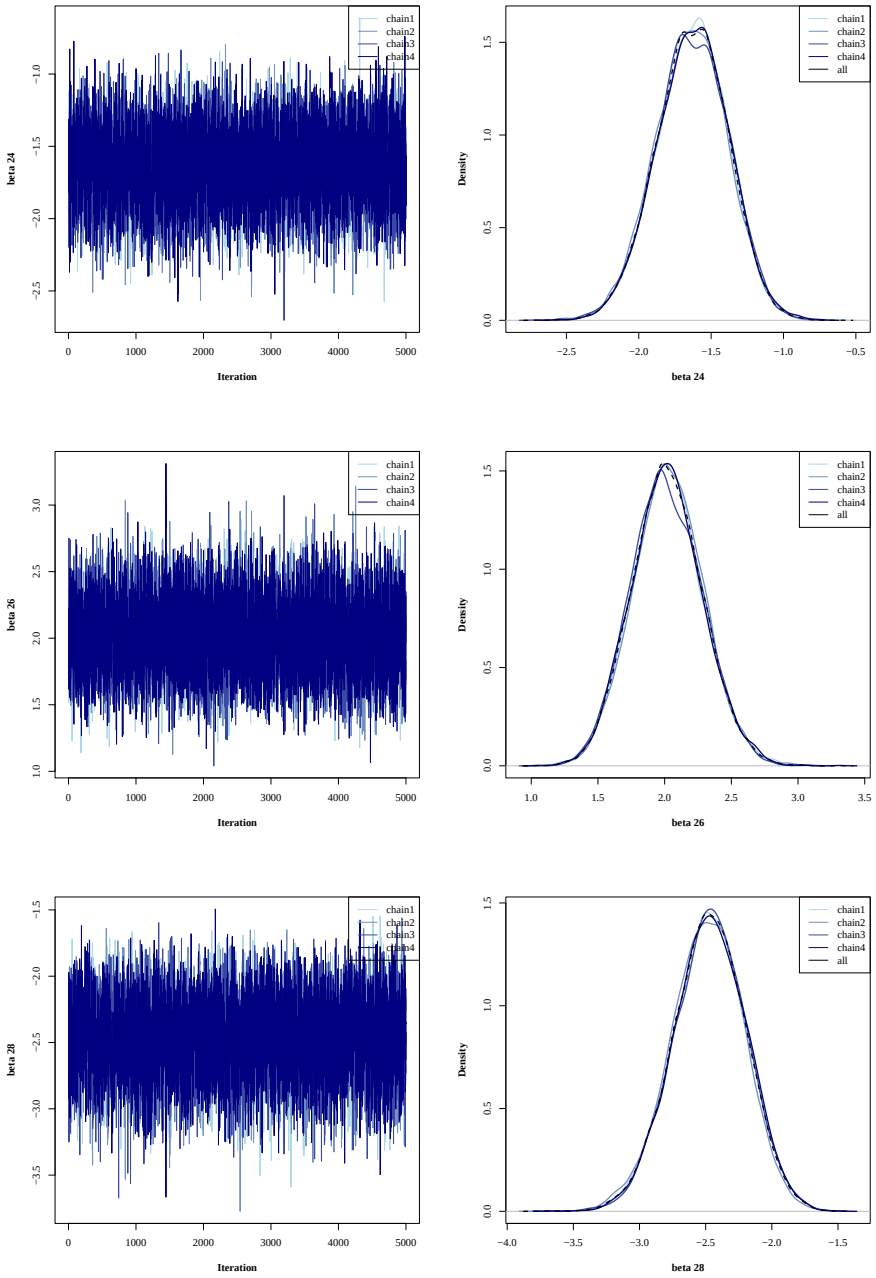


Figure 6: Trace plot (left) and density plot (right) for β^*_{24}, β^*_{26} and β^*_{28} in the case of 6 non-zero coefficients of the Cox-RHS prior with a Weibull baseline function, while $n = 200$, $p = 150$. The black dashed line in the density plot represents the case of all samples.

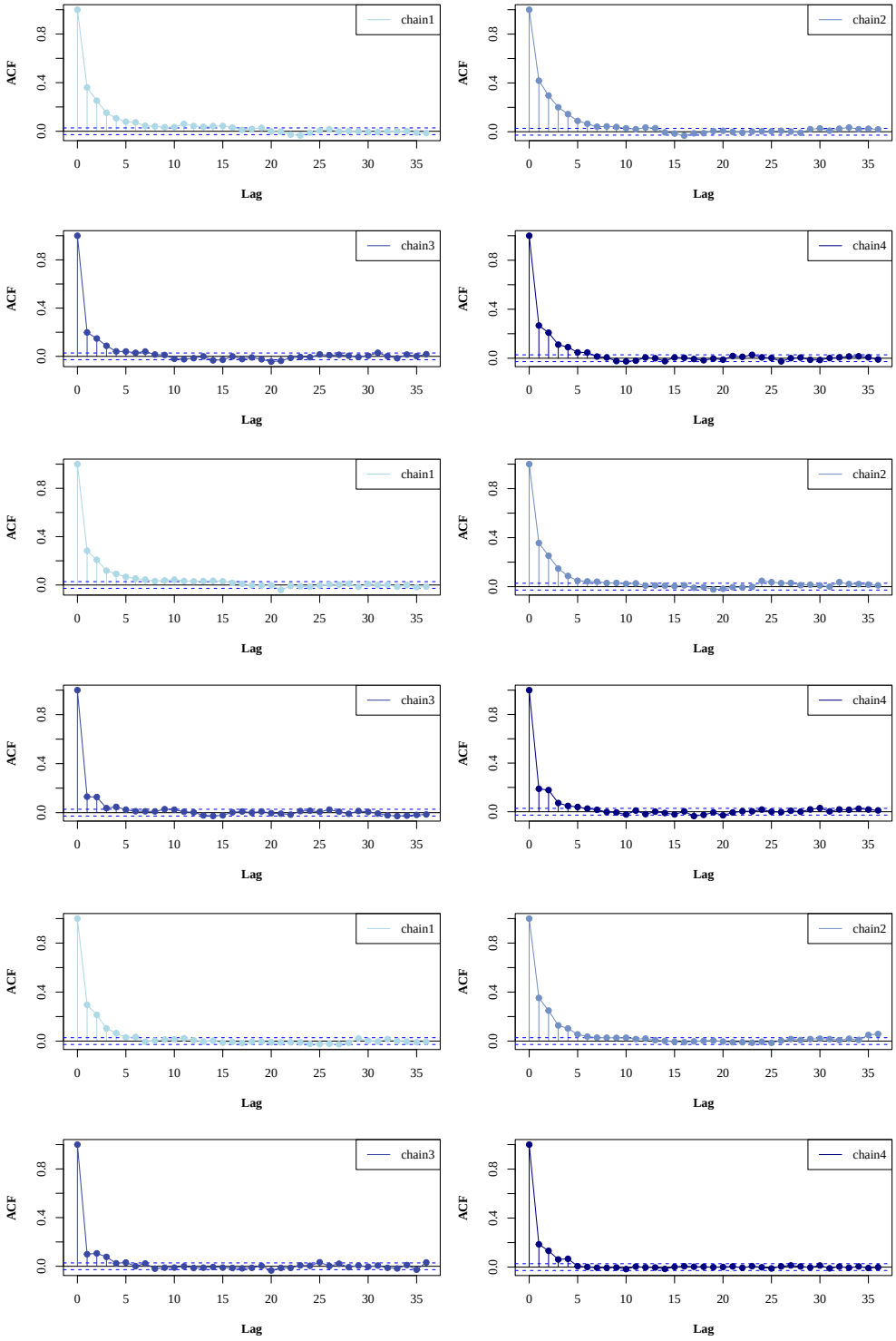


Figure 7: ACF plot for β^*_{18}, β^*_{20} and β^*_{22} in the case of 6 non-zero coefficient of the Cox-RHS prior with a Weibull baseline function while $n = 200$, $p = 150$.

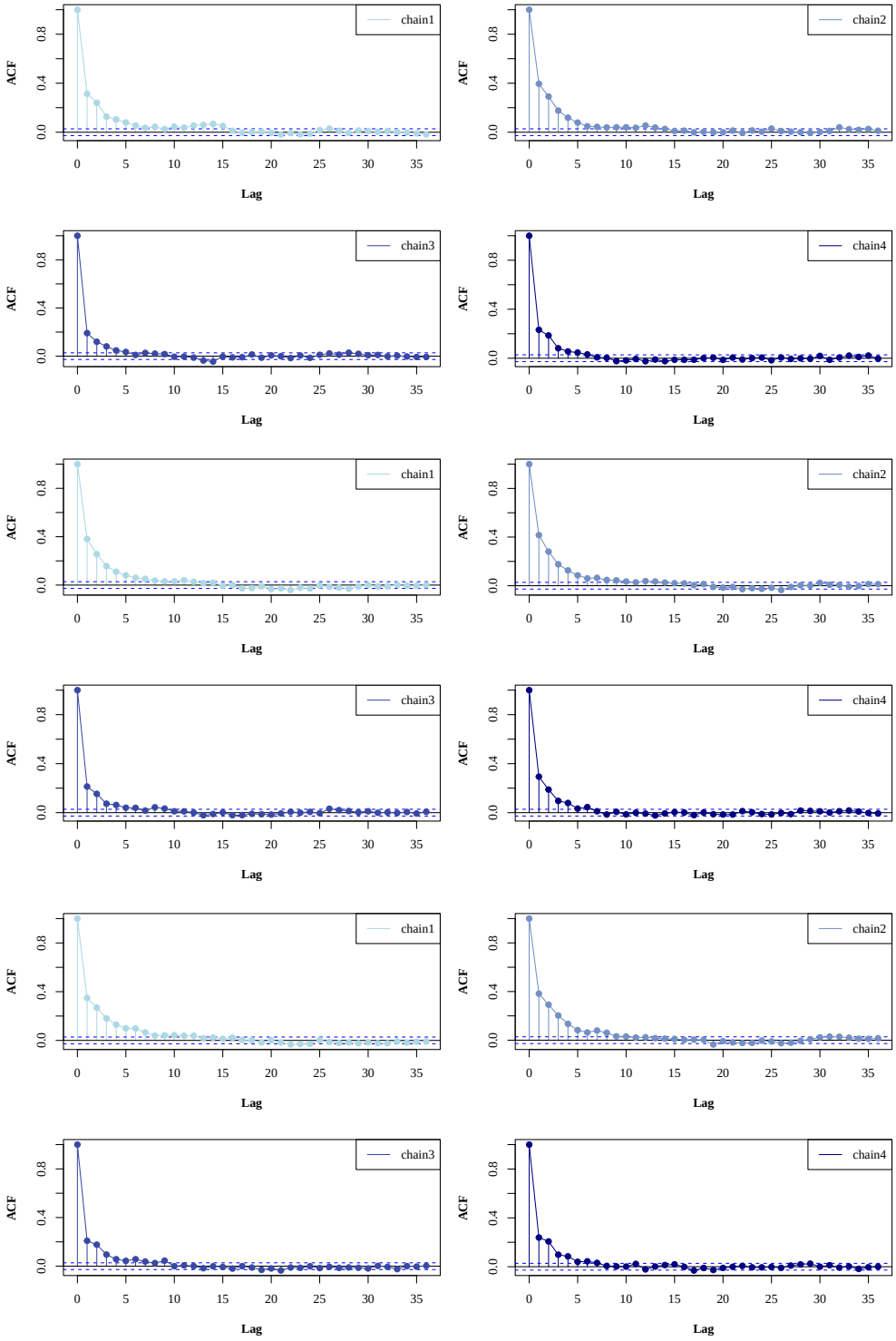


Figure 8: ACF plot for β^*_{24}, β^*_{26} and β^*_{28} in the case of 6 non-zero coefficient of the Cox-RHS prior with a Weibull baseline function while $n = 200$, $p = 150$.

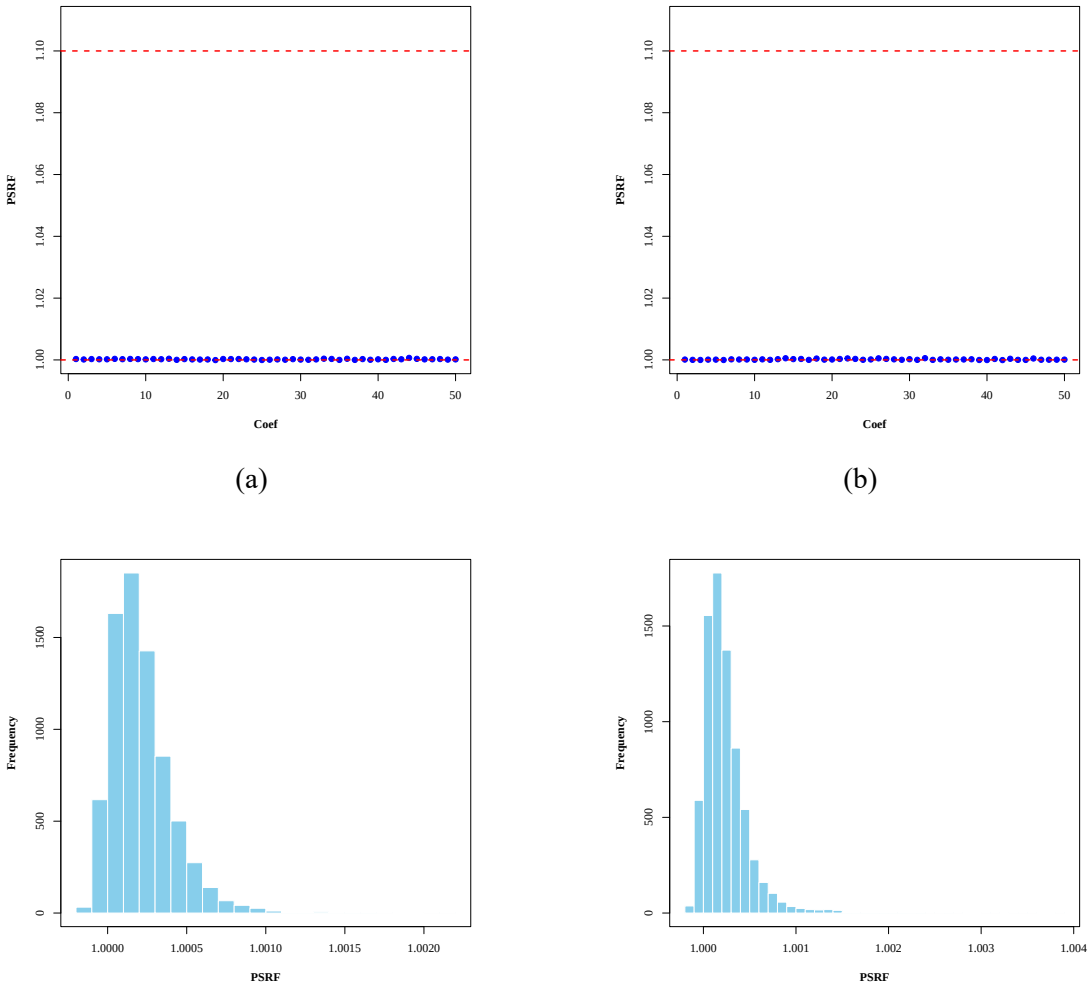


Figure 9: Diagnostics of $\hat{R}$ for replicates 1 and 50 of Cox-RHS under Weibull baseline function with $n = 200$. The scatter plots diagnose for $p = 50$ and the histograms for $p = 150$. (a) Scatter plot of $\hat{R}$ in replicate 1 for 6 non-zero coefficient setup. (b) Scatter plot of $\hat{R}$ in replicate 1 for 12 non-zero coefficient setup. (c) Histogram of $\hat{R}$ of 50 replicates for 6 non-zero coefficient setup. (d) Histogram of $\hat{R}$ of 50 replicates for 12 non-zero coefficient setup. Additionally, the red dashed lines in (a) and (b) indicate the thresholds 1 and 1.1.

C Supplementary Results of Real Data Application

C.1 Coefficient Estimated by Posterior Median and MAP

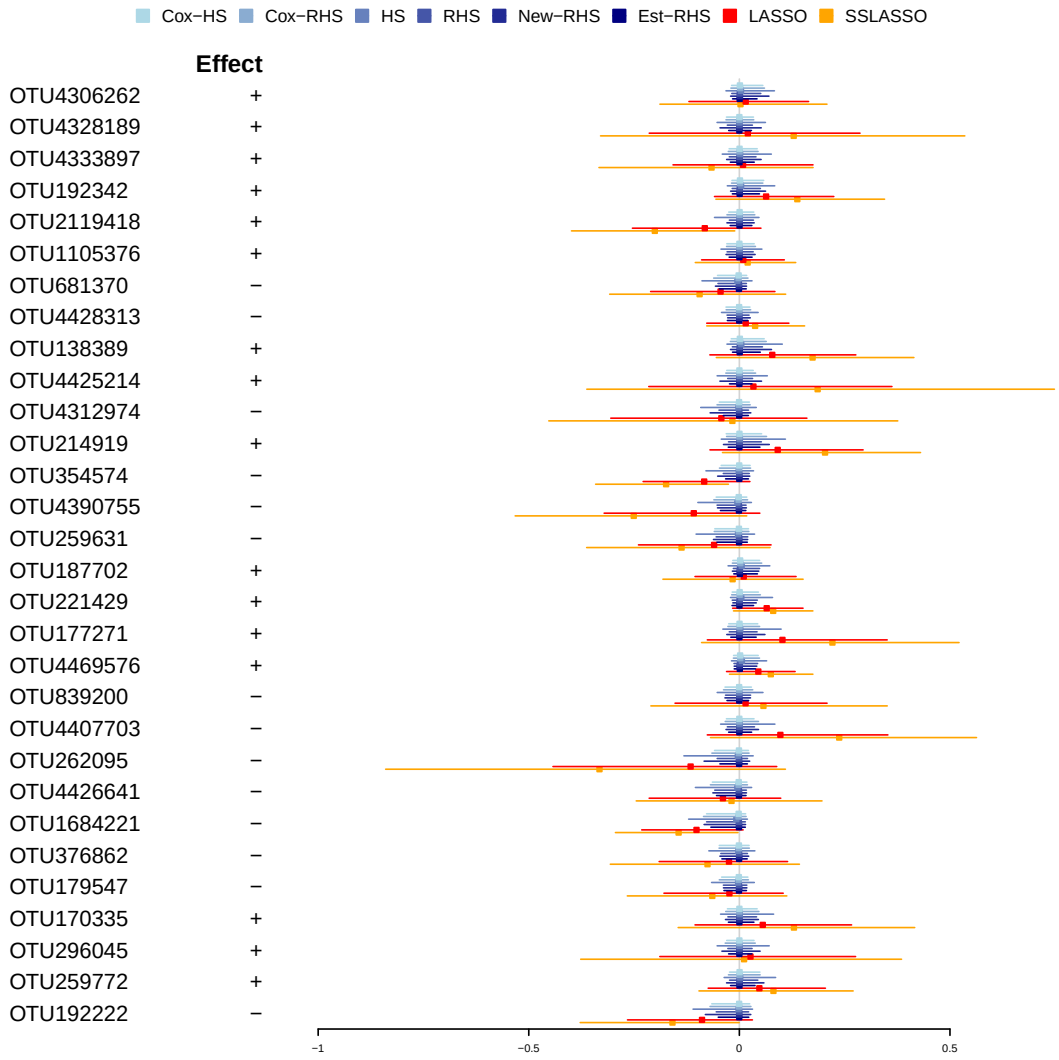


Figure 10: Coefficient estimated by posterior median for each compositional variable.

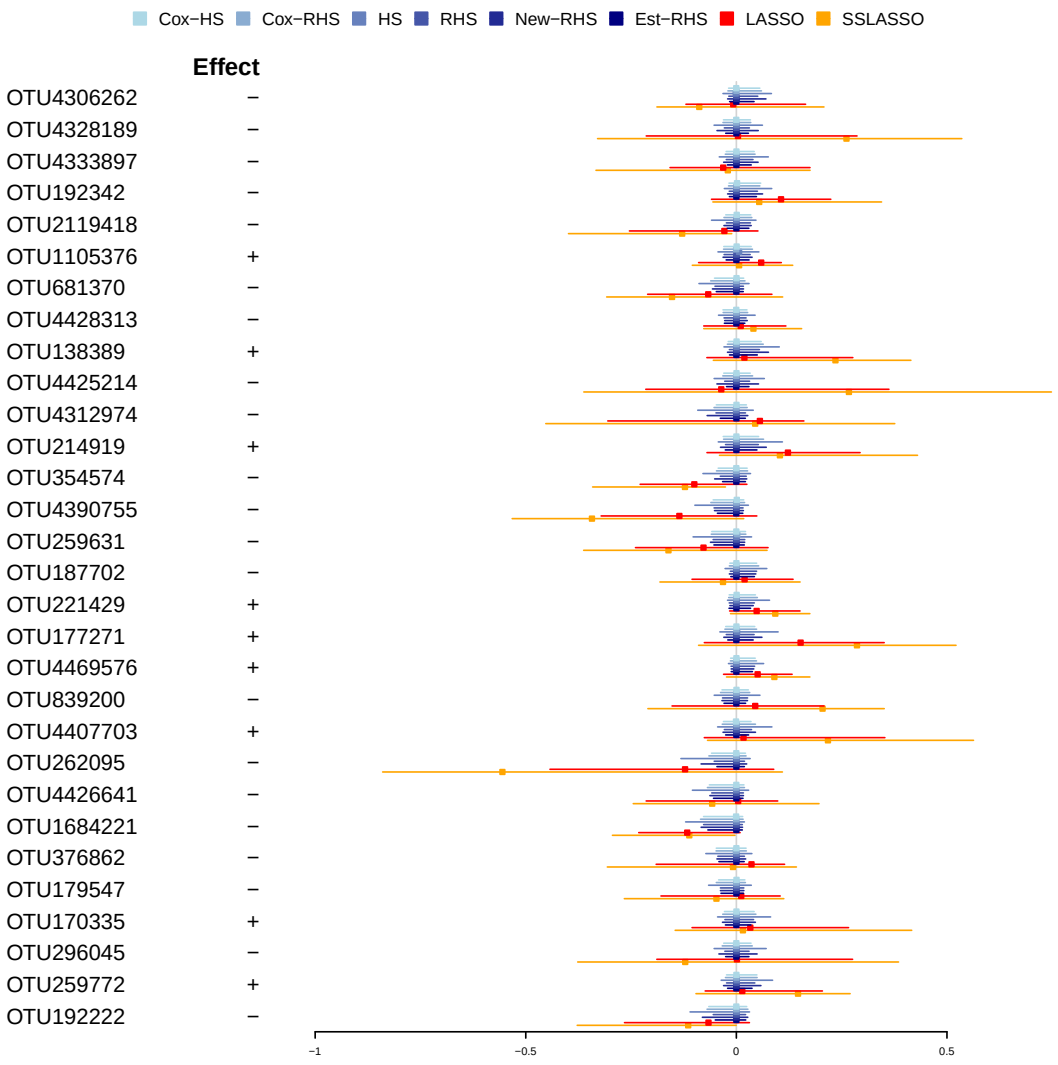


Figure 11: Coefficient estimated by MAP for each compositional variable.